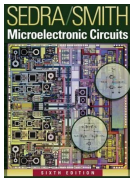
电子科技大学
University of Electronic Science and Technology of China



Fundamentals of Analogue Circuits
模拟电路基础

电子科技大学格拉斯哥学院

BAO JINGFU

鲍 景富

baojingfu@uestc.edu.cn

Introduction to this Course

- **Professor: BAO JINGFU**
  - 1.Email: baojingfu@uestc.edu.cn.
  - 2.Office: A512 （IC_Building）
- **TEXT: Microelectronic Circuits,**
  Sedra/Smith, Oxford Publishing

Do you have difficulty learning?

"If one cannot light a light bulb with a battery and wire then everything built on those ideas has problems."

baojingfu@uestc.edu.cn

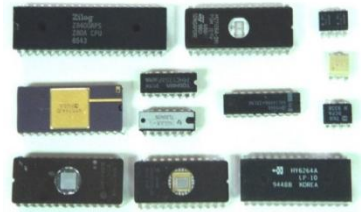
Methods to improve Learning

- Active Learning
- flipping the Traditional Lecture
- Case/Project-based
- Internet

baojingfu@uestc.edu.cn

Development of electronics depends on the development of electronic components.

Vaccum Tube→Transistor→Integrated Circuit

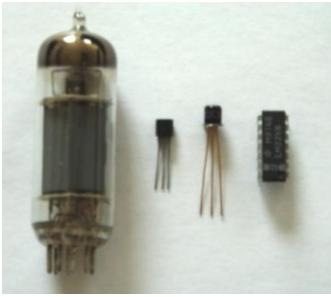


Yr.1904
Vaccum Tube！！

Yr.1947
Transistor

Yr.1958
Integrated Circuits(IC)

The development of electronics also promoted the development of electronic components.



Comparison of volume

baojingfu@uestc.edu.cn

Development of Semiconductor Components

- **Yr.1947 First Transistor**
- **Yr.1958 First IC**
- **Yr.1969 LSI**
- **Yr.1975 VLSI**

First IC: Integrated ONLY Four Transistors

Yr.1997. Four billion transistors

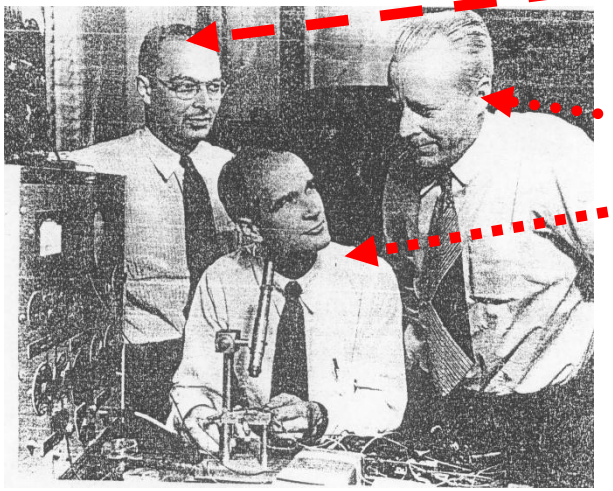
Integration increased by 10 times/6 years

Where is the limit? (Why?)。

So the electronic technology is still in rapid development!

baojingfu@uestc.edu.cn

Some Scientist should be memorized.



The inventor of the first transitor

（by John Bardeen , William Schockley and Walter Brattain in Bell Lab）

November,1947. Named it "transistor" on 16,December,1947.

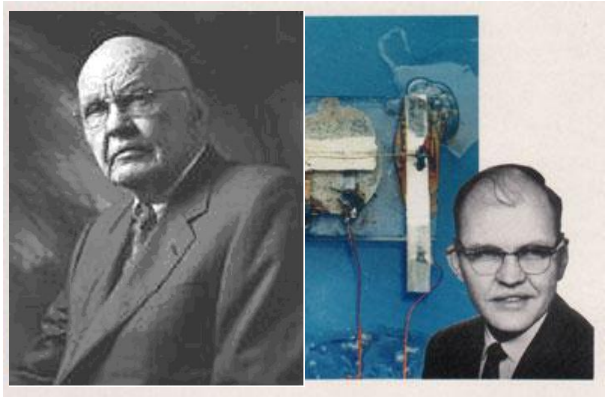
Nobel Prize on 1956

First IC and its inventor

（ Jack Kilby from TI ）

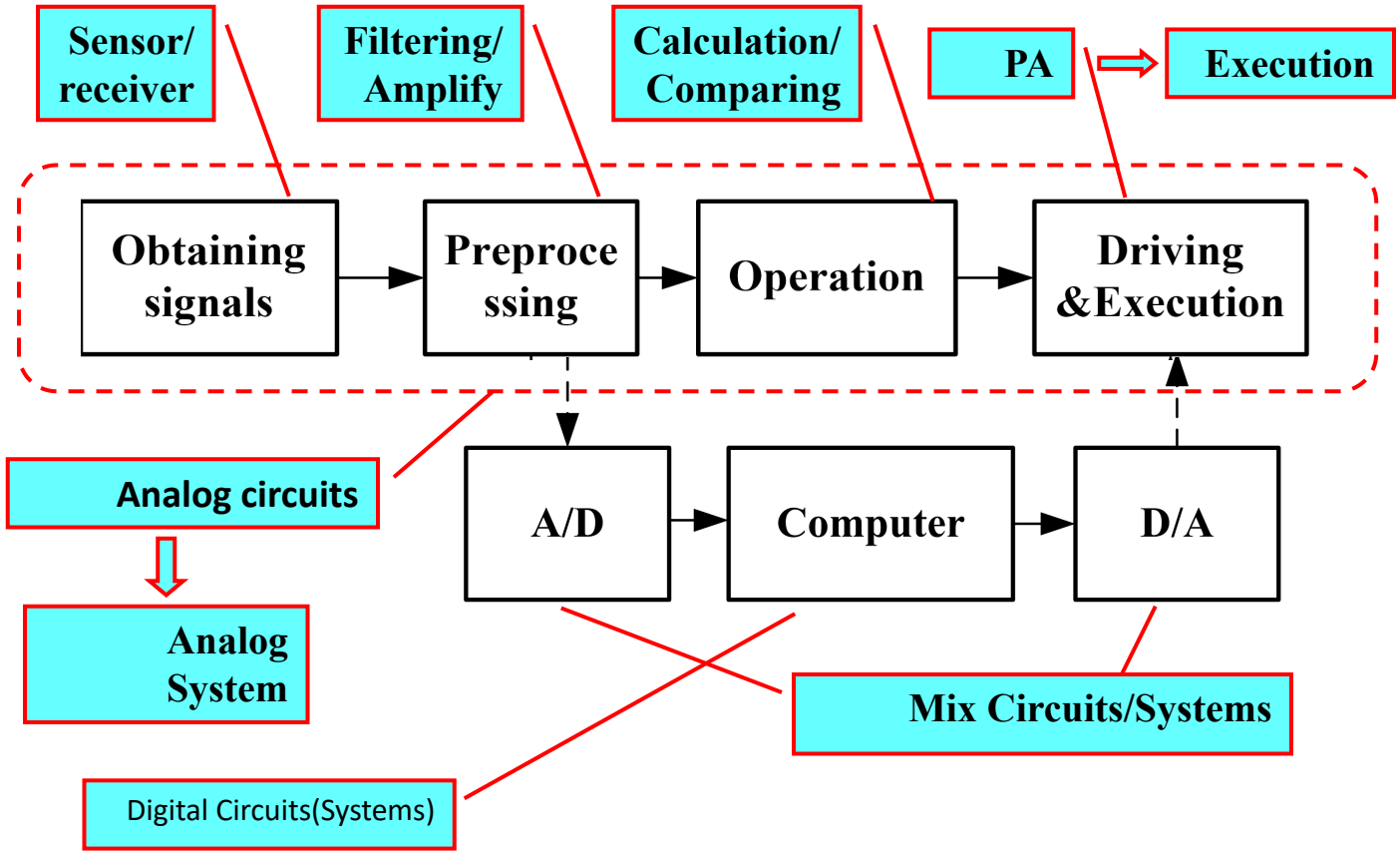
1958.9.12

Yr.2000: Nobel Prize。



baojingfu@uestc.edu.cn

Block Diagram of Electronic System



baojingfu@uestc.edu.cn

course outcome

Theory and Analysis

a1. Describe the physical operation of a diode and explain the VCR for a diode.

a2. Describe the basic action of a BJT/MOSFET including its characteristic curves.

a3. Apply standard parametric models of a BJT/ MOSFET transistor.

a4. Analyse and design standard circuits to bias a transistor.

a5. Describe what is meant by the frequency response of an amplifier circuits.

a6. Explain the specifications of a differential amplifier.

a7. Describe the basic types of power amplifier, including classes A, B and AB, and compare their operation critically

a8. Analyse general circuits with negative feedback, including the four topologies and dual-port network models

baojingfu@uestc.edu.cn

course outcome

Design and Practice and Experiments

b1. Measure the characteristic curves of BJT and MOSFET transistors respectively.

b2. Design differential amplifiers and measure the DC/AC/Trans performances.

b3. Design multistage amplifiers and measure the performances.

b4. Design negative feedback amplifiers and measure the performances.

b5. Design, implement and test a single transistor amplifier circuit.

b6. Design and implement a two-stage amplifier circuit.

b7. Design and implement a negative feedback amplifier circuit using integrated operational amplifier.

Ability to Communicate Effectively

c1. Demonstrate ability to prepare a technical report relating the lab experiment, including a discussion comparing experimental measurements with theory and simulation.

Use of Computer Tools

k1. Apply circuit CAD tools used to each of the design tasks in Outcomes b1 to b4 above.

baojingfu@uestc.edu.cn

How to learn this course

1. Engineering, Practice, Application

- Based on application requirements.
- Analysis should be verified by practice.

2. Basic Concepts & Circuits & Analysis Method.

- Basic Concepts: one concepts—different applications.
- Basic Circuits: one principle, different Circuits.
- Basic Method: depends on different type of circuits to analysis different specifications.

3. Qualitative vs Quantitative analysis

4. Chose the appropriate circuits/Weigh the pros and cons

Common laws: KCL, KVL, Superposition,

Thevenin's theorem, Norton's theorem

baojingfu@uestc.edu.cn

Overall course grading policies

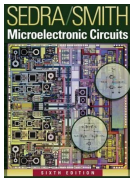
- 10% homework and in-class records
- 15% labs and term project
- 75% open-book final exam (2 hours)

baojingfu@uestc.edu.cn

Some important rules

1. **Copying or plagiarism of any works (e.g. homework\report) is a violation Code of Academic Integrity and can results "0" grade of the course.**
2. **Generally, be present yourself in the lab every time for experiments or project. Or you will be given "0" of the experiments or project.**
3. **Follow the rules in lab.**

baojingfu@uestc.edu.cn



Chapter #1: Signals and Amplifiers

from **Microelectronic Circuits** Text
by Sedra and Smith
Oxford Publishing

baojingfu@uestc.edu.cn

Introduction

- **IN THIS CHAPTER YOU WILL LEARN…**
  - That electronic circuits process signals, and thus understanding electrical signals is essential to appreciating the material in this book.
  - The Thevenin and Norton representations of signal sources.
  - The representation of a signal as sum of sine waves.
  - The analog and digital representations of a signal.

baojingfu@uestc.edu.cn

Introduction

- **IN THIS CHAPTER YOU WILL LEARN…**
  - The most basic and pervasive signal-processing function: signal amplification, and correspondingly, the signal amplifier.
  - How amplifiers are characterized (modeled) as circuit building blocks independent of their internal circuitry.
  - How the frequency response of an amplifier is measured

baojingfu@uestc.edu.cn

1.1. Signals

- **signal** – contains information
  - e.g. voice of radio announcer reading the news
- **process** – an operation which allows an observer to understand this information from a signal
  - generally done electrically
- **transducer** – device which converts signal from non-electrical to electrical form
  - e.g. microphone (sound to electrical)

baojingfu@uestc.edu.cn

1.1: Signals

- **Q:** How are signals represented?
  - **A: Thevenin form** – voltage source $\mathbf{v}_s(t)$ with series resistance R_S
    - preferable when R_S is low
  - **A: Norton form** – current source $\mathbf{i}_s(t)$ with parallel resistance R_S
    - preferable when R_S is high

baojingfu@uestc.edu.cn

1.1. Signals

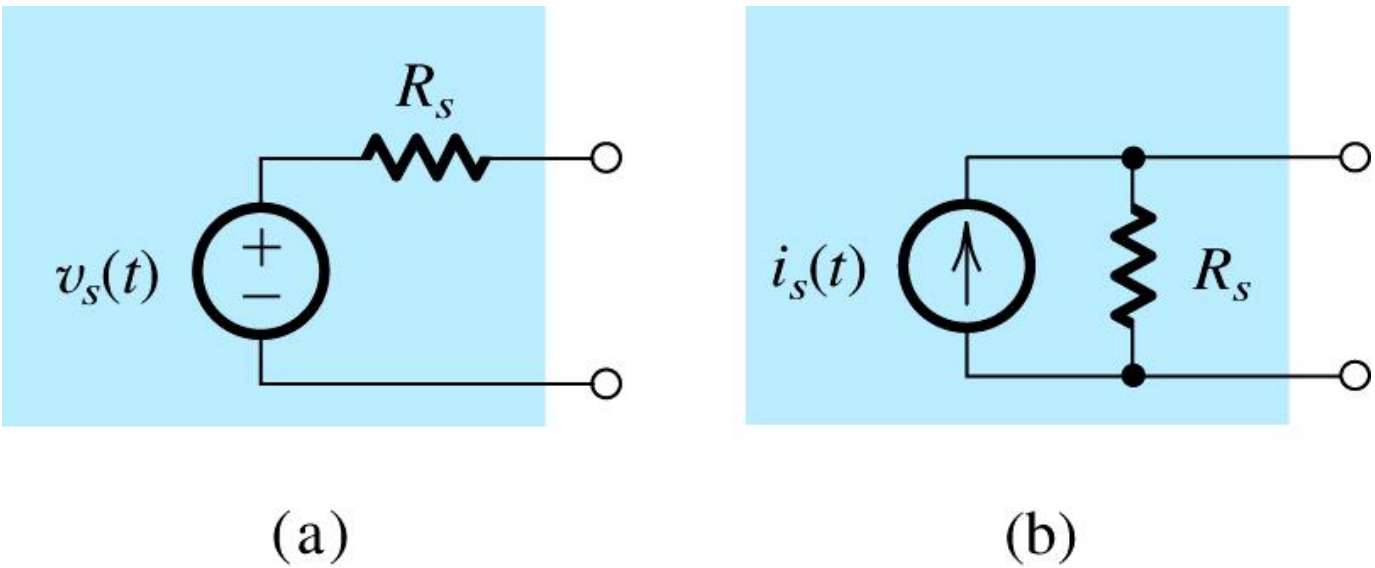


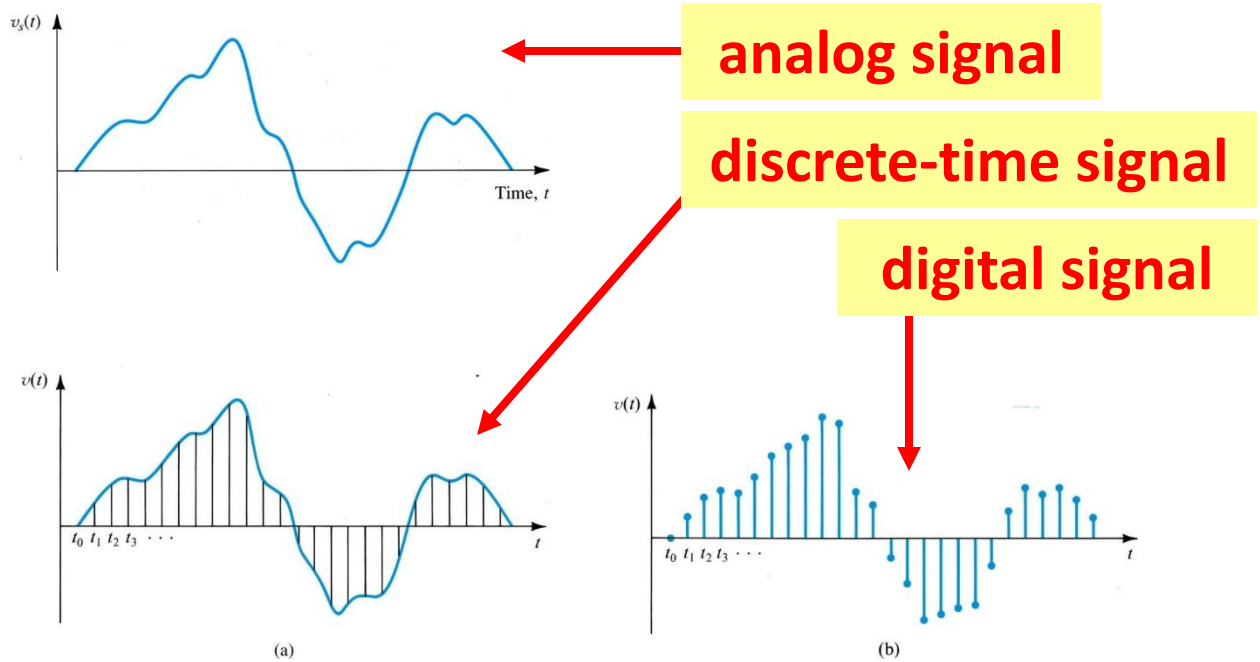
Figure 1.1: Two alternative representations of a signal source: **(a)** the Thévenin form; **(b)** the Norton form.

1.2. Analog and Digital Signals

- **analog signal** – is continuous with respect to both value and time
- **discrete-time signal** – is continuous with respect to value but sampled at discrete points in time
- **digital signal** – is quantized (applied to values) as well as sampled at discrete points in time

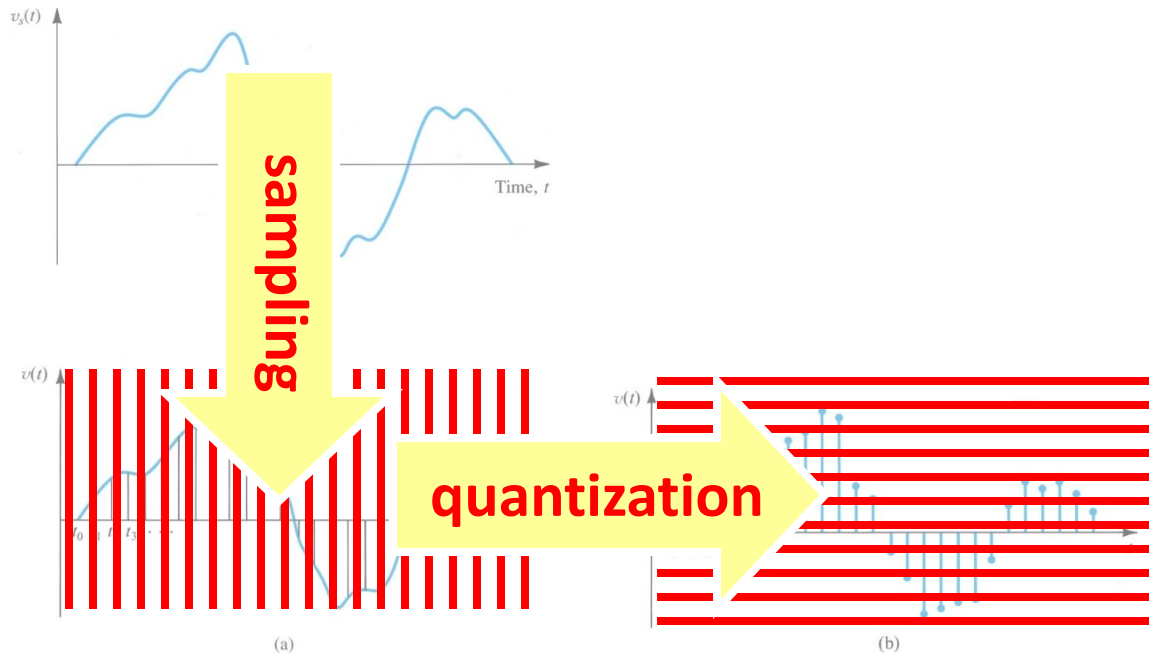
baojingfu@uestc.edu.cn

1.2. Analog and Digital Signals



baojingfu@uestc.edu.cn

1.2. Analog and Digital Signals



baojingfu@uestc.edu.cn

1.3. Frequency Spectrum of Signals

- **frequency spectrum** – defines a time-domain signal in terms of the strength of harmonic components
  - **Q:** What is a Fourier Series?
    - **A:** An expression of a periodic function as the sum of an infinite number of sinusoids whose frequencies are harmonically related

baojingfu@uestc.edu.cn

What is a Fourier Series?

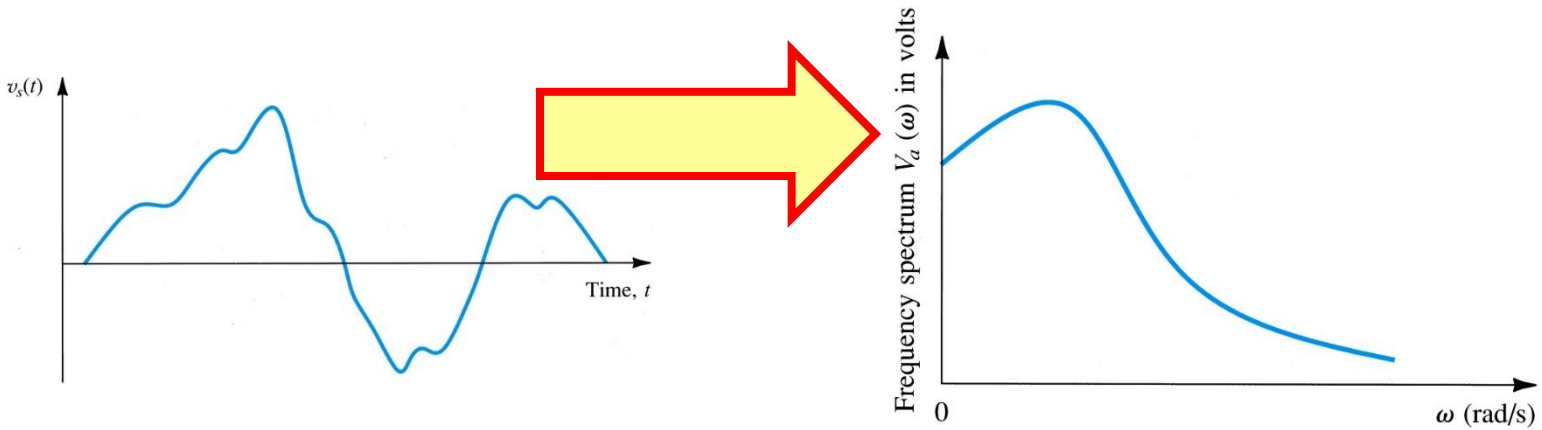
- **decomposition** – of a periodic function into the (possibly infinite) sum of simpler oscillating functions

$$\overbrace{\mathbf{f}(x) = \frac{a_0}{2} + \underbrace{\sum_{k=1}^{\infty} [a_k \mathbf{cos}(kx) + b_k \mathbf{sin}(kx)]}_{\substack{a_k = \frac{1}{\pi}\int_{-\pi}^{\pi} \mathbf{f}(x)\mathbf{cos}(kx)dx\,,\ n\geq 0 \\ b_k = \frac{1}{\pi}\int_{-\pi}^{\pi} \mathbf{f}(x)\mathbf{sin}(kx)dx\,,\ n\geq 1}}}^{\text{Fourier Series Representation of } \mathbf{f}(x)}$$

baojingfu@uestc.edu.cn

1.3. Frequency Spectrum of Signals

- **Q:** Can the Fourier Transform be applied to a non-periodic function of time?
  - **A:** Yes, however (as opposed to a discrete frequency spectrum) it will yield a continuous…



1.4. Amplifiers

- **Q:** Why is signal amplification needed?
  - **A:** Because many transducers yield output at low power levels (*mW*)
- **linearity** – is property of an amplifier which ensures a signal is not "altered" from amplification
- **distortion** – is any unintended change in output

baojingfu@uestc.edu.cn

1.4.1. Signal Amplification

- **voltage amplifier** – is used to boost voltage levels for increased resolution.
- **power amplifier** – is used to boost current levels for increased "intensity".

$$\overbrace{\mathbf{v}_o(t) = A_v \cdot \mathbf{v}_i(t)}^{\text{output / input relationship for amplifier}}$$

voltage gain

baojingfu@uestc.edu.cn

1.4.2. Amplifier Circuit Symbol

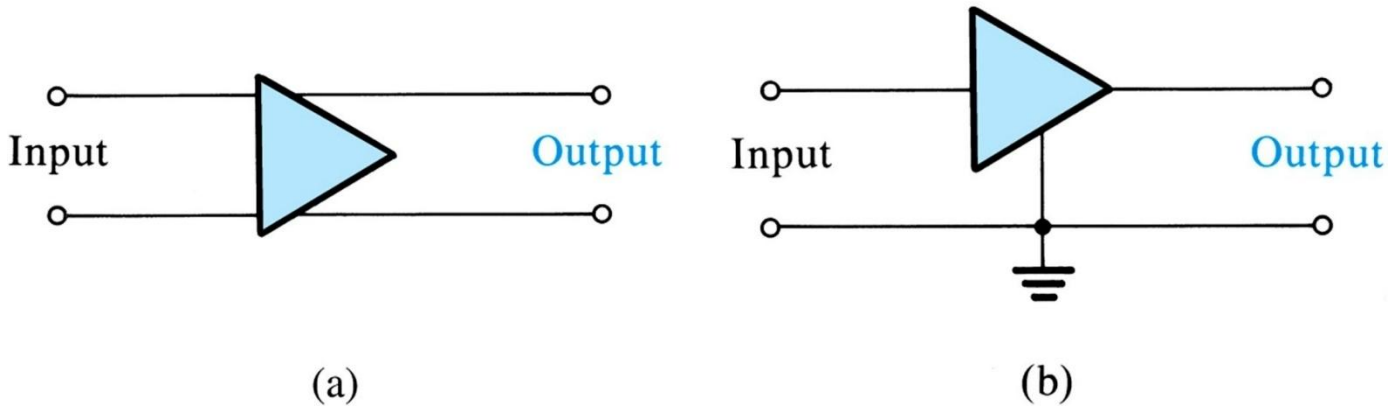


Figure 1.11: (a) Circuit symbol for amplifier. **(b)** An amplifier with a common terminal (ground) between the input and output ports.

1.4.4. Power and Current Gain

- **Q:** What is one main difference between an amplifier and transformer? …Because both alter voltage levels.
  - **A:** Amplifier may be used to boost power delivery.

$$power\ gain\ (A_p) = \frac{load\ power\ (P_L)}{input\ power\ (P_I)} = \frac{v_o i_o}{v_i i_i}$$

baojingfu@uestc.edu.cn

1.4.5. Expressing Gain in Decibels

- **Q:** How may gain be expressed in **decibels**?

$$\text{voltage gain in decibels} = 20\,\mathbf{log}|A_v|dB$$

$$\text{current gain in decibels} = 20\,\mathbf{log}|A_i|dB$$

$$\text{power gain in decibels} = 10\,\mathbf{log}(A_p)dB$$

baojingfu@uestc.edu.cn

1.5. Circuit Models for Amplifiers

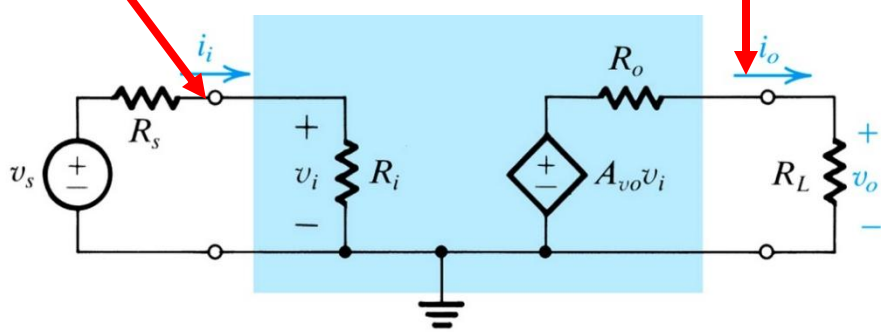
- **model** – is the description of component's (e.g. amplifier) terminal behavior
  - neglecting internal operation / transistor design

baojingfu@uestc.edu.cn

1.5.1. Voltage Amplifiers

$$\overbrace{\text{input voltage} = v_i = \underbrace{(v_s)}_{\text{source volt.}} \frac{R_i}{\underbrace{R_i + R_s}_{\text{source and input resistances}}}}^{\text{model of amplifier input terminals}}$$

$$\overbrace{\text{output voltage} = v_o = \underbrace{(A_{vo} v_i)}_{\text{open-ckt output voltage}} \frac{R_L}{\underbrace{R_L + R_o}_{\text{output and load resistances}}}}^{\text{model of amplifier output terminals}}$$



(b)

baojingfu@uestc.edu.cn

1.5.1. Voltage Amplifiers

- **Q:** How can one model the amplifier behavior mathematically from previous slide?
  - **A:** Model which is function of: v_s, A_{vo}, R_i, R_s, R_o, R_L

$$v_o = \underbrace{\left(A_{vo} \underbrace{(v_s)}_{\text{source volt.}} \underbrace{\frac{R_i}{R_i + R_s}}_{\text{source and input resistances}} \right)}_{\text{open-ckt output voltage}} \underbrace{\frac{R_L}{R_L + R_o}}_{\text{output and load resistances}}$$

$$= A_{vo} v_s \frac{R_i}{R_i + R_s} \frac{R_L}{R_L + R_o}$$

baojingfu@uestc.edu.cn

1.5.1. Voltage Amplifiers

- **Q:** What is one "problem" with this behavior?
  - **A:** Gain (ratio of v_o and v_s) is not constant, and dependent on input and load resistance.

$$v_o = \underbrace{\left(A_{vo} \underbrace{(v_s)}_{\text{source volt.}} \underbrace{\frac{R_i}{R_i + R_s}}_{\text{source and input resistances}} \right)}_{\text{open-ckt output voltage}} \underbrace{\frac{R_L}{R_L + R_o}}_{\text{output and load resistances}}$$

$$= A_{vo} v_s \frac{R_i}{R_i + R_s} \frac{R_L}{R_L + R_o}$$

The **ideal amplifier model** neglects this nonlinearity.

baojingfu@uestc.edu.cn

1.5.1. Voltage Amplifiers

- **ideal amplifier model** – is function of v_s and A_{vo} only!!
  - It is assumed that $R_o << R_L$…
  - It is assumed that $R_s << R_i$ …

$$v_o = \underbrace{A_{vo} v_s \frac{R_i}{R_i + R_s} \frac{R_L}{R_L + R_o}}_{\text{non-ideal model}} = \underbrace{A_{vo} v_s}_{\text{ideal model}}$$

key characteristics of ideal voltage amplifier model = source resistance R_S and load resistance R_L have no effect on gain

baojingfu@uestc.edu.cn

1.5.2. Cascaded Amplifiers

- In real life, an amplifier is not ideal and will not have infinite input impedance or zero output impedance.
- **Cascading** of amplifiers, however, may be used to emphasize desirable characteristics.
  - first amplifier – high R_i, medium R_o
  - last amplifier – medium R_i, low R_o
  - aggregate – high R_i, low R_o

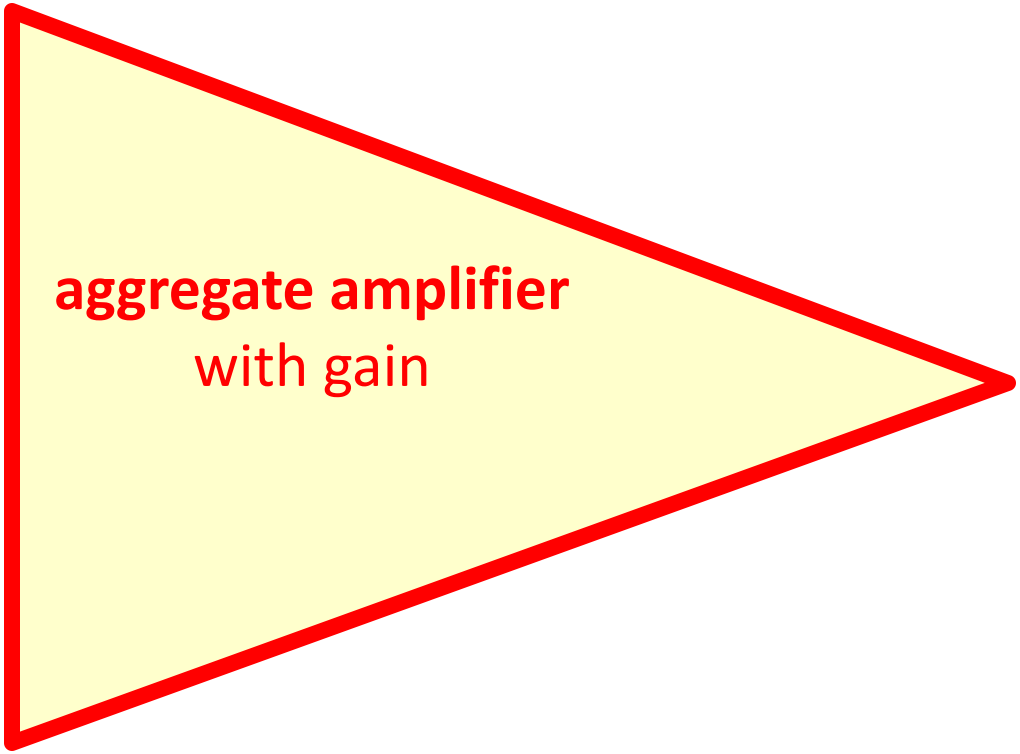
baojingfu@uestc.edu.cn

Example 1.3: Cascaded Amplifier Configurations

- Examine system of cascaded amplifiers on next slide.
- **Q(a):** What is overall voltage gain?
- **Q(b):** What is overall current gain?
- **Q(c):** What is overall power gain?

baojingfu@uestc.edu.cn

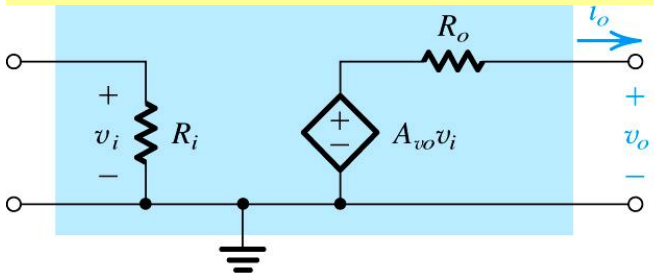
Example 1.3: Cascaded Amplifier Configurations



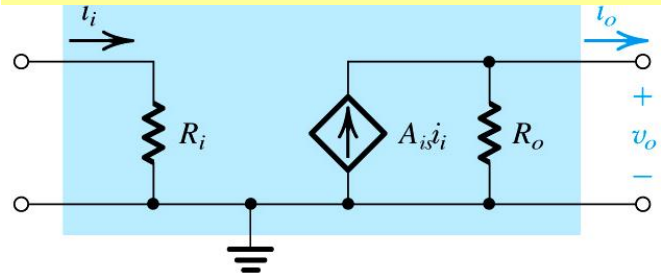
baojingfu@uestc.edu.cn

1.5.3. Other Amplifier Types

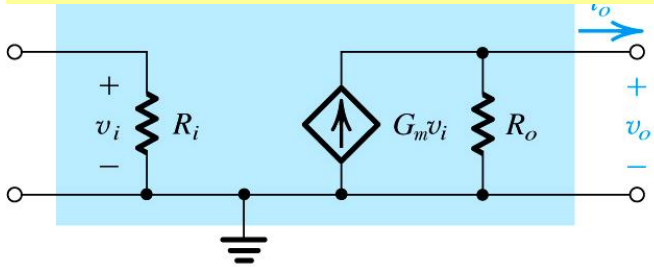
voltage amplifier



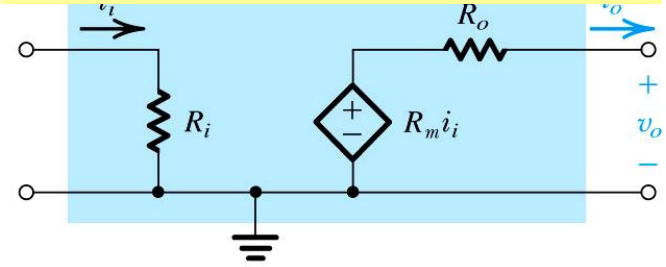
current amplifier



transconductance amp.



transresistance amp.

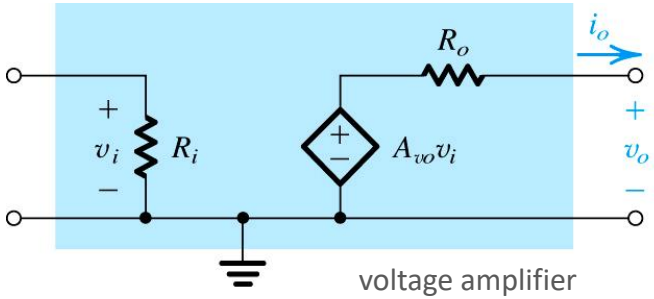


baojingfu@uestc.edu.cn

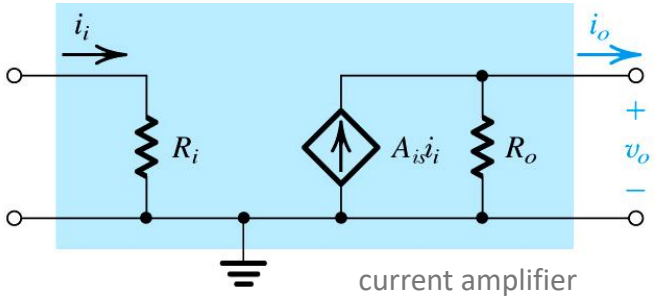
1.5.3. Other Amplifier Types

$$A_{v0} = \left.\frac{v_0}{v_i}\right|_{i_0=0} \quad \text{with} \quad R_i = \infty, \; R_o = 0$$

$$A_{is} = \left.\frac{i_0}{i_i}\right|_{v_0=0} \quad \text{with} \quad R_i = 0, \; R_o = \infty$$

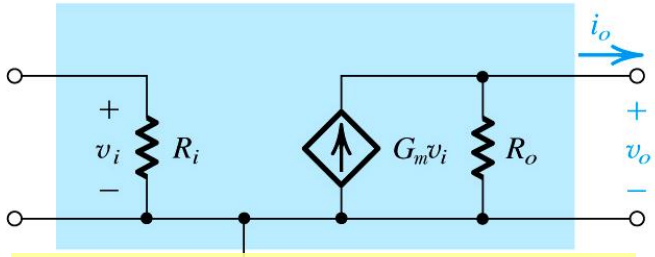


voltage amplifier

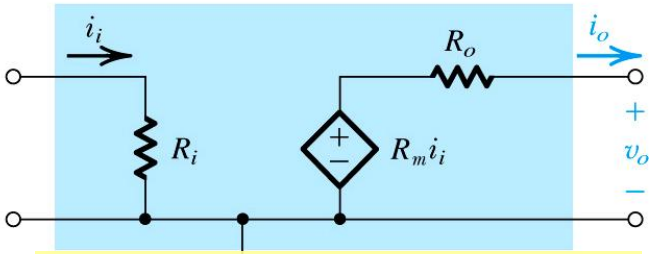


current amplifier

transconductance amplifier



transresistance amplifier



$$G_m = \left.\frac{i_0}{v_i}\right|_{v_0=0} \quad \text{with} \quad R_i = \infty, \; R_o = \infty$$

$$R_m = \left.\frac{v_0}{i_i}\right|_{i_0=0} \quad \text{with} \quad R_i = 0, \; R_o = 0$$

baojingfu@uestc.edu.cn

1.5.4. Relationship Between Four Amp Models

- **interchangeability** –

  although these four types exist, any of the four may be used to model any amplifier

  –they are related through A_{vo} (open circuit gain)

$$A_{vo} = \overbrace{A_{is}\left(\frac{R_o}{R_i}\right)}^{\text{current to voltage amplifier}} = \overbrace{G_m R_o}^{\text{transcond. to voltage amplifier}} = \overbrace{\frac{R_m}{R_i}}^{\text{transres. to voltage amplifier}}$$

baojingfu@uestc.edu.cn

1.5.5. Determining R_i and R_o

- **Q:** How can one calculate input resistance from terminal behavior?
  - **A:** Observe v_i and i_i, calculate via $R_i = v_i / i_i$
- **Q:** How can one calculate output resistance from terminal behavior?
  - **A:**
    - Remove source voltage (such that $v_i = i_i = 0$)
    - Apply voltage to output (v_x)
    - Measure negative output current ($-i_o$)
    - Calculate via $R_o = -v_x / i_o$

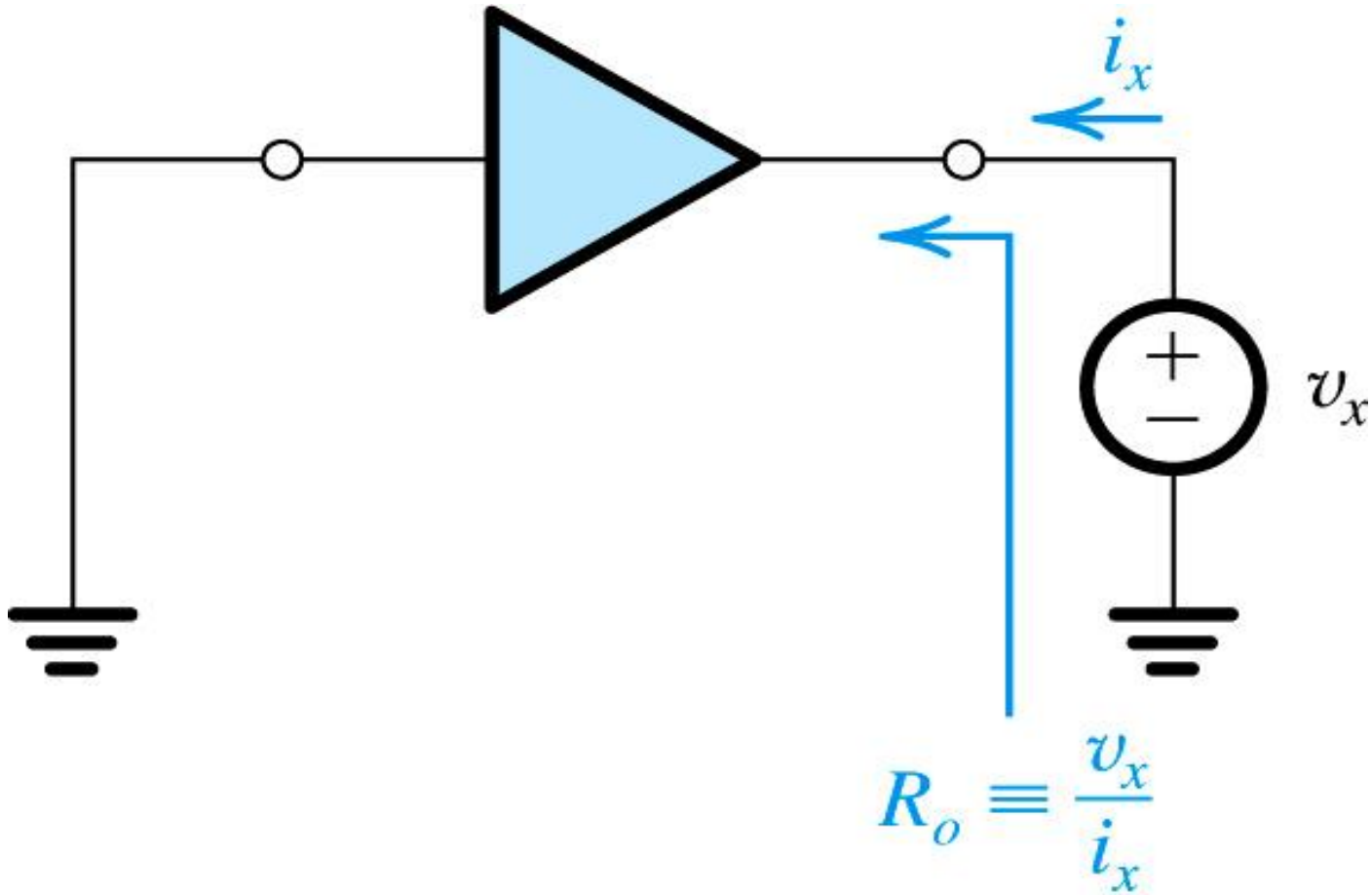


Figure 1.18: Determining the output resistance.

Example 1.5

Figure 1.25 shows a voltage amplifier having an input resistance R_i, an input capacitance C_i, a gain factor μ, and an output resistance R_o. The amplifier is fed with a voltage source V_s having a source resistance R_s, and a load of resistance R_L is connected to the output.

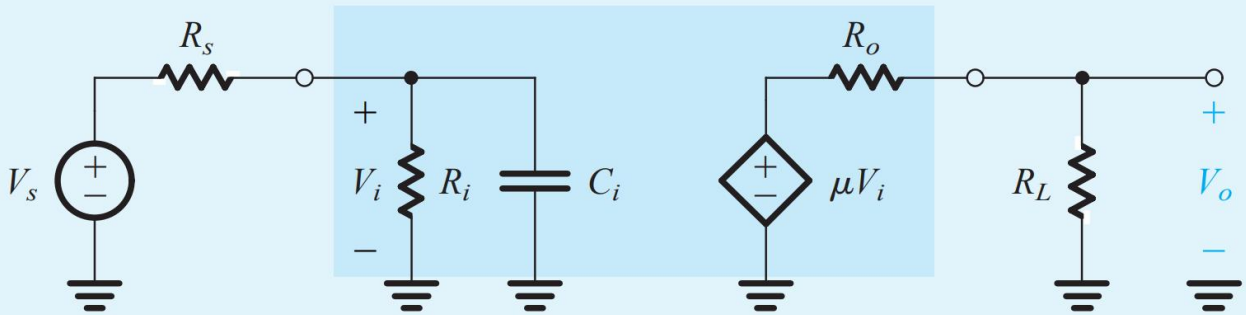


Figure 1.25 Circuit for Example 1.5.

(a) Derive an expression for the amplifier voltage gain V_o/V_s as a function of frequency. From this find expressions for the dc gain and the 3-dB frequency.

(b) Calculate the values of the dc gain, the 3-dB frequency, and the frequency at which the gain becomes 0 dB (i.e., unity) for the case $R_s = 20$ kΩ, $R_i = 100$ kΩ, $C_i = 60$ pF, $\mu = 144$ V/V, $R_o = 200$ Ω, and $R_L = 1$ kΩ.

(c) Find $v_o(t)$ for each of the following inputs:

(i) $v_i = 0.1 \sin 10^2\, t$, V

(ii) $v_i = 0.1 \sin 10^5\, t$, V

(iii) $v_i = 0.1 \sin 10^6\, t$, V

(iv) $v_i = 0.1 \sin 10^8\, t$, V

Solution

(a) Utilizing the voltage-divider rule, we can express V_i in terms of V_s as follows

$$V_i = V_s \frac{Z_i}{Z_i + R_s}$$

where Z_i is the amplifier input impedance. Since Z_i is composed of two parallel elements, it is obviously easier to work in terms of $Y_i = 1/Z_i$. Toward that end we divide the numerator and denominator by Z_i, thus obtaining

$$V_i = V_s \frac{1}{1 + R_s Y_i}$$

$$= V_s \frac{1}{1 + R_s[(1/R_i) + sC_i]}$$

Thus,

$$\frac{V_i}{V_s} = \frac{1}{1 + (R_s/R_i) + sC_i R_s}$$

This expression can be put in the standard form for a low-pass STC network (see the top line of Table 1.2) by extracting $[1 + (R_s/R_i)]$ from the denominator; thus we have

$$\frac{V_i}{V_s} = \frac{1}{1 + (R_s/R_i)} \frac{1}{1 + sC_i[(R_s R_i)/(R_s + R_i)]} \tag{1.20}$$

At the output side of the amplifier we can use the voltage-divider rule to write

$$V_o = \mu V_i \frac{R_L}{R_L + R_o}$$

baojingfu@uestc.edu.cn

This equation can be combined with Eq. (1.20) to obtain the amplifier transfer function as

$$\frac{V_o}{V_s} = \mu \frac{1}{1+(R_s/R_i)} \frac{1}{1+(R_o/R_L)} \frac{1}{1+sC_i[(R_sR_i)/(R_s+R_i)]} \tag{1.21}$$

We note that only the last factor in this expression is new (compared with the expression derived in the last section). This factor is a result of the input capacitance C_i, with the time constant being

$$\begin{aligned} \tau &= C_i \frac{R_sR_i}{R_s+R_i} \\ &= C_i(R_s \parallel R_i) \end{aligned} \tag{1.22}$$

We could have obtained this result by inspection: From Fig. 1.25 we see that the input circuit is an STC network and that its time constant can be found by reducing V_s to zero, with the result that the resistance seen by C_i is R_i in parallel with R_s. The transfer function in Eq. (1.21) is of the form $K/(1+(s/\omega_0))$, which corresponds to a low-pass STC network. The dc gain is found as

$$K \equiv \frac{V_o}{V_s}(s=0) = \mu \frac{1}{1+(R_s/R_i)} \frac{1}{1+(R_o/R_L)} \tag{1.23}$$

The 3-dB frequency ω_0 can be found from

$$\omega_0 = \frac{1}{\tau} = \frac{1}{C_i(R_s \parallel R_i)} \tag{1.24}$$

baojingfu@uestc.edu.cn

Since the frequency response of this amplifier is of the low-pass STC type, the Bode plots for the gain magnitude and phase will take the form shown in Fig. 1.23, where K is given by Eq. (1.23) and ω_0 is given by Eq. (1.24).

(b) Substituting the numerical values given into Eq. (1.23) results in

$$K = 144 \frac{1}{1+(20/100)} \frac{1}{1+(200/1000)} = 100 \text{ V/V}$$

Thus the amplifier has a dc gain of 40 dB. Substituting the numerical values into Eq. (1.24) gives the 3-dB frequency

$$\omega_0 = \frac{1}{60 \text{ pF} \times (20 \text{ k}\Omega // 100 \text{ k}\Omega)}$$

$$= \frac{1}{60 \times 10^{-12} \times (20 \times 100/(20+100)) \times 10^3} = 10^6 \text{ rad/s}$$

Example 1.5 *continued*

Thus,

$$f_0 = \frac{10^6}{2\pi} = 159.2 \text{ kHz}$$

Since the gain falls off at the rate of –20 dB/decade, starting at ω_0 (see Fig. 1.23a) the gain will reach 0 dB in two decades (a factor of 100); thus we have

$$\text{Unity-gain frequency} = 100 \times \omega_0 = 10^8 \text{ rad/s or } 15.92 \text{ MHz}$$

baojingfu@uestc.edu.cn

(c) To find $v_o(t)$ we need to determine the gain magnitude and phase at 10^2, 10^5, 10^6, and 10^8 rad/s. This can be done either approximately utilizing the Bode plots of Fig. 1.23 or exactly utilizing the expression for the amplifier transfer function,

$$T(j\omega) \equiv \frac{V_o}{V_s}(j\omega) = \frac{100}{1 + j(\omega/10^6)}$$

We shall do both:

(i) For $\omega = 10^2$ rad/s, which is $(\omega_0/10^4)$, the Bode plots of Fig. 1.23 suggest that $|T| = K = 100$ and $\phi = 0°$. The transfer function expression gives $|T| \simeq 100$ and $\phi = -\tan^{-1} 10^{-4} \simeq 0°$. Thus,

$$v_o(t) = 10 \sin 10^2 t, \text{ V}$$

(ii) For $\omega = 10^5$ rad/s, which is $(\omega_0/10)$, the Bode plots of Fig. 1.23 suggest that $|T| \simeq K = 100$ and $\phi = -5.7°$. The transfer function expression gives $|T| = 99.5$ and $\phi = -\tan^{-1} 0.1 = -5.7°$. Thus,

$$v_o(t) = 9.95 \sin(10^5 t - 5.7°), \text{ V}$$

(iii) For $\omega = 10^6$ rad/s $= \omega_0$, $|T| = 100/\sqrt{2} = 70.7$ V/V or 37 dB and $\phi = -45°$. Thus,

$$v_o(t) = 7.07 \sin(10^6 t - 45°), \text{ V}$$

(iv) For $\omega = 10^8$ rad/s, which is $(100\omega_0)$, the Bode plots suggest that $|T| = 1$ and $\phi = -90°$. The transfer function expression gives

$$|T| \simeq 1 \quad \text{and} \quad \phi = -\tan^{-1} 100 = -89.4°$$

Thus,

$$v_o(t) = 0.1 \sin(10^8 t - 89.4°), \text{ V}$$

Conclusion

- An electrical signal source can be represented in either Thevenin form (a voltage source v_s in series with source resistance R_s) or the Norton form (a current source i_s in parallel with resistance R_s). The Thevenin voltage v_s is the open-circuit voltage between the source terminals. The Norton current i_s is equal to the short-circuit current between the source terminals. For the two representations to be equivalent, v_s and $R_s i_s$ must be equal.
- A signal can be represented either by its waveform vs time or as the sum of sinusoids. The latter representation is known as the frequency spectrum of the signal.

Conclusion (2)

- Analog signals have magnitudes that can assume any value. Electronic circuits that process analog signals are called analog circuits. Sampling the magnitude of an analog signal at discrete instants of time and representing each signal sample by a number results in a digital signal. Digital signals are processed by digital circuits.

baojingfu@uestc.edu.cn

Conclusion (3)

- Amplifiers increase the signal power and thus require dc power supplies for their operation.
- The amplifier voltage gain can be expressed as a ratio A_v in Vo/Vin or in decibels, $20\mathbf{log}|A_v|$ in dB.
- Depending on the signal to be amplified (voltage or current) and on the desired form of output signal (voltage or current) there are four basic amplifier types: voltage, current, transconductance, and transresistance. A given amplifier may be modeled by any of these configurations, in which case their parameters are related by (1.14) through (1.16) in the text.

Homework

1.43 Consider the voltage-amplifier circuit model shown in Fig. 1.16(b), in which $A_{vo} = 10$ V/V under the following conditions:

(a) $R_i = 10R_s$, $R_L = 10R_o$
(b) $R_i = R_s$, $R_L = R_o$
(c) $R_i = R_s/10$, $R_L = R_o/10$

1.44 An amplifier with 40 dB of small-signal, open-circuit voltage gain, an input resistance of 1 MΩ, and an output resistance of 10 Ω, drives a load of 100 Ω. What voltage and power gains (expressed in dB) would you expect with the load connected? If the amplifier has a peak output-current limitation of 100 mA, what is the rms value of the largest sine-wave input for which an undistorted output is possible? What is the corresponding output power available?

Homework(2)

1.48 You are given two amplifiers, A and B, to connect in cascade between a 10-mV, 100-kΩ source and a 100-Ω load. The amplifiers have voltage gain, input resistance, and output resistance as follows: for A, 100 V/V, 10 kΩ, 10 kΩ, respectively; for B, 1 V/V, 100 kΩ, 100 Ω, respectively. Your problem is to decide how the amplifiers should be connected. To proceed, evaluate the two possible connections between source S and load L, namely, SABL and SBAL. Find the voltage gain for each both as a ratio and in decibels. Which amplifier arrangement is best?

baojingfu@uestc.edu.cn

Homework(3)

1.53 A current amplifier for which $R_i = 1\ \text{k}\Omega$, $R_o = 10\ \text{k}\Omega$, and $A_{is} = 100$ A/A is to be connected between a 100-mV source with a resistance of 100 kΩ and a load of 1 kΩ. What are the values of current gain i_o/i_i, of voltage gain v_o/v_s, and of power gain expressed directly and in decibels?

1.54 A transconductance amplifier with $R_i = 2\ \text{k}\Omega$, $G_m = 40$ mA/V, and $R_o = 20\ \text{k}\Omega$ is fed with a voltage source having a source resistance of 2 kΩ and is loaded with a 1-kΩ resistance. Find the voltage gain realized.

Homework(4)

D 1.57 It is required to design an amplifier to sense the open-circuit output voltage of a transducer and to provide a proportional voltage across a load resistor. The equivalent source resistance of the transducer is specified to vary in the range of 1 kΩ to 10 kΩ. Also, the load resistance varies in the range of 1 kΩ to 10 kΩ. The change in load voltage corresponding to the specified change in R_s should be 10% at most. Similarly, the change in load voltage corresponding to the specified change in R_L should be limited to 10%. Also, corresponding to a 10-mV transducer open-circuit output voltage, the amplifier should provide a minimum of 1 V across the load. What type of amplifier is required? Sketch its circuit model, and specify the values of its parameters. Specify appropriate values for R_i and R_o of the form 1×10^m Ω.

Thanks so much

baojingfu@uestc.edu.cn